Hazard Identification and Inspection Protocol Sheet

Log Transport (CU732044)

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Log Transport (CU732044)

Description of Work

The Log Transport classification unit involves the transportation of logs from the cut–block to a log dump, dry land sort or secondary processing facility such as a mill by way of a logging truck. The following sections provide a description of the work that occurs.

Planning

During the planning phase, employers are responsible for ensuring truck drivers are adequately trained and have the appropriate credentials (e.g., air brake ticket, class 1, etc.). Employers are also responsible for ensuring that drivers receive orientation on the following topics:

- Weather conditions
  - Prime contractors and sub–contractors must ensure that, prior to arriving on site, all truck drivers are aware of the site specific wet weather shut down thresholds (water balance as determined by rain gauge readings)
    - This includes how and when the site will be shut down, who will communicate this information, and how it will be communicated
- Road use hazards, additional traffic and flow of traffic
- Location and grade of steep haul roads
- Ensuring hauling direction is suitable to existing road design
- Workplace rules and regulations for log landings, dry land sorts and mill yards
- Pick up/drop off locations

Site Prep

Truck drivers are responsible for inspecting logging truck equipment daily including communication systems prior to the use or travel on a resource road(new requirement of 26.65 requires daily inspection of the cab guards and mounting system). Employers are also responsible for identifying all steep haul roads and ensuring risk assessments are conducted by a qualified person and ensuring the log truck drivers have been informed of the location and the hazards of the steep haul roads and the information, recommendations and instruction from the risk assessment (see OHSR 26.2 Guideline), and ensuring they have a qualified man check system and making themselves aware of the forest fire control plans.

Loading Logs

If the logging truck is a conventional logging truck, meaning no self-loading capabilities, then the logs are loaded onto the truck by a log loader. If the logging truck is self-loading log truck then the truck driver in most cases loads the logs onto the truck. The hazards and controls depend then on the method of log loading.

Transporting

Transporting the logs occurs in a variety of weather conditions, road type and grade (steepness of road) all of which affect the hazards and controls used.

Unloading

As with log loading unloading can occur using a log loader if it is a conventional logging truck or by the truck driver if the truck is a self-loader. The hazards and controls depend then on the method used to unload the logs.

Reloading Trailer

Reloading of the trailer onto the truck can occur via a number of ways (log transporter trailer hoists, mobile equipment, pre-load logging trucks). This depends on the unique operating characteristics of the worksite.

Truck Maintenance

This section includes the regular maintenance of truck body, cab guards and tires while following proper lockout procedures for truck maintenance and proper procedures for tire servicing.

Legislation

The following sections of the OHSR capture the primary activities included in the Log Transport classification unit:

- OHSR Section 26.2, Planning and conducting a forestry operation
- WCA Sections 21–25

Hazards, Controls and Legislation

Phase	Hazard	Control	Legislation
Site Prep – Driver Orientation	• Inadequate orientation could result in driver not having clear instructions on truck configuration and workplace safety	• Drivers are required to have a site specific orientation	WCA Section 21, *General duties of employers* OHSR Section 3.23, *Young or new worker orientation and training* OHSR Section 26.2, *Planning and conducting a forestry operation* OHSR Section 26.5, *Initial safety meeting*
	• Inappropriate or non–use of seatbelt	• Drivers are required by regulation and the law to wear seatbelts at all times	OHSR Section 16.32, *Provision* OHSR Section 16.33, *Use* *Motor Vehicle Act*
Site Prep – Inspection of truck equipment	• Unprepared for the work or upset condition	• Pre–trip inspections	OHSR Section 4.3, *Safe machinery and equipment* OHSR Section 16.34, *Start of shift inspection* OHSR Section 26.71.2, *Daily log* WCA Section 21, *General duties of employers*
	• Truck backing up without alerting people in the vicinity	• Truck is equipped with a backup alarm and backup lights	OHSR Section 16.8, *Warning signal device – backup alarm* OHSR Section 16.9, *Lights*

Phase	Hazard	Control	Legislation
	• In the truck, the operator does not have clear vision	• Windshields, side and rear windows, and rear–vision mirrors must be maintained	OHSR Section 16.12, *Maintenance*
	• Not maintaining a secure operators cab • In the event of a collision or rollover drivers can be struck by loose tools or equipment	• Maintain clean and secure cab	OHSR Section 16.35, *Securing tools and equipment* OHSR Section 17.5, *Securing equipment*
	• Welding grab hooks on cinches • This practice can fatigue and weaken the cinch • Operators can be struck by the cinch when they break	• Do not weld on cinches • There are now manufactured cinches with grab hooks installed	OHSR Section 4.3(1)(i), *Safe machinery and equipment*
Site Prep – Communication	• Truck is not equipped with the correct radio and/or radio channels for the roads being accessed	• Ensuring radios are functioning (radio check) • Ensuring radio channels are correct for the roads being accessed (radio check)	OHSR Section 26.2, *Planning and conducting a forestry operation*

Phase	Hazard	Control	Legislation
Site Prep – Identifying steep haul roads	• Not identifying locations of potential hazards for drivers	• Employers expected to be knowledgeable of all known hazards of the road system	WCA Section 25, General duties of owners WCA Section 21, General duties of employers WCA Section 24, Coordination of multiple-employer workplaces OHSR Section 26.2, Planning and conducting a forestry operation
Site Prep – Person check system	• Workers working alone	• Develop person check system that will ensure drivers well being	OHSR Section 4.20.2, Hazard identification, elimination and control OHSR Section 4.21, Procedures for checking well-being of worker
Site Prep – Forest fire awareness and preparedness	• Unprepared workers in the event of a forest fire	• Ensure workers are trained in the S100 training	OHSR Section 26.3.1, Forestry operation fire fighting
Loading – Self Loading Logging Trucks	• Drivers working alone and they in equipment that is unguarded	• Person check system	OHSR Section 4.21, Procedures for checking well-being of worker
		• Safe work procedures (guarding)	WCA Section 21, General duties of employers OHSR Section 26.13.1(1), Equipment operator protections
		• Safe guards on the loading equipment (boom deflectors, etc.)	OHSR Section 26.13.1(2), Equipment operator protections

Phase	Hazard	Control	Legislation
		• Log deck piles should be safe for the self-loading operators (i.e., should not exceed the height of the operators feet)	OHSR Section 26.2, *Planning and conducting a forestry operation*
	• Maintenance within self loaders often done at heights over 10 feet (log loader)	• Fall protection • Lockout procedures	Safe work procedures WCA Section 21, *General duties of employers*
	• Loss of stability of the tractor	• Ensuring outriggers are placed as per manufacturer's instructions and that they are well maintained and safe for use	OHSR Section 4.3, *Safe machinery and equipment*
Loading – Conventional Logging Trucks	• Exposure to mobile equipment	• Driver orientation to the hazards of the landing	OHSR Section 26.5, *Initial safety meeting*
	• Operator's position during loading single or multiple bunks	• Operator must not stand on the cab platform, or between the cab and a log being loaded	OHSR Section 26.67, *Load specifications*
	• Suspended logs being passed over the driver and/or the driver's cab	• Safe work procedures and safe landing procedures	WCA Section 21, *General duties of employers* OHSR Section 26.59, *Suspended logs*

Phase	Hazard	Control	Legislation
	• Operator climbing on bunks to raise or lower stakes	• Operator must not go on bunks and trailer assemblies	OHSR Section 26.66(4.1), *Bunks and stakes*
		• If the operator must go on a bunk, they must be provided with a safe means to do so (i.e. a non-slip stationary work platform 20” wide, and the means for three points for contact)	OHSR Section 26.66(4.2), *Bunks and stakes*
	• Exposure to unstable logs on the load while attempting to secure the load with binders	• Proper loading procedures and properly built loads (i.e. loads must be stable without the use of binders) • Ensure safe work procedures are in place (as a number of recent injuries have occurred during this work process) • The logs of the bottom tier and side rows must exceed beyond the front and rear bunks and stakes, according to the regulation	OHSR Section 26.67, *Load specifications*

Phase	Hazard	Control	Legislation
	• Significant MSI injuries from the application of binders without mechanical assistance (drivers throwing the binders over the load)	• Unless it is not possible, use equipment or an alternate means of throwing the binder or pulling the binder over the load (using a ball and strawline rope)	OHSR Section 26.68, *Binders* Safe work procedures OHSR Section 4.47, *Risk identification* OHSR Section 4.48, *Risk assessment* OHSR Section 4.51, *Education and training* OHSR Section 4.52, *Evaluation*
	• Temporary transport without binders (short duration to hammer)	• Binders being applied to the load prior to leaving the landing and or the loader	OHSR Section 26.68, *Binders*
	• Lack of communication between the truck driver and the loader operator	• Proper signaling as per regulation and/or two way radio communication	OHSR Section 26.67(1), *Load specifications*
Transporting	• Loss of control due to physical impairment (hours of work)	• Supervision	WCA Section 23, *General duties of supervisors*
		• Hours of work regulated for drivers by Provincial Legislation	OHSR Section 4.19, *Physical or mental impairment*
		• Log books required	OHSR Section 26.71.2

Phase	Hazard	Control	Legislation
	• Loss of control due to substance abuse	• Supervision	WCA Section 23, *General duties of supervisors*
		• Worker responsibility	WCA Section 22, *General duties of workers* OHSR Section 4.20, *Impairment by alcohol, drug or other substance*
	• Loss of control due to speed	• Supervision	WCA Section 23, *General duties of supervisors*
		• Worker responsibility	WCA Section 22, *General duties of workers* OHSR Section 4.20, *Impairment by alcohol, drug or other substance*
		• Motor Vehicle Act	No Jurisdiction
		• Forestry Act	No jurisdiction
	• Loss of control due to weather	• Supervision	WCA Section 23, *General duties of supervisors* OHSR Section 26.2, *Planning and conducting a forestry operation*
		• Worker responsibility to use proper equipment (i.e. chains)	WCA Section 22, *General duties of workers*

Phase	Hazard	Control	Legislation
	• Loss of control due to steep haul roads	• Employer required to conduct site specific risk assessments for steep haul roads • Refer to OHSR 26.2 Guideline for further information	OHSR Section 26.2, *Planning and conducting a forestry operation*
	• Loss of control due to mechanical failure	• Pre-trip inspections and timely maintenance and brake checks as required for the road	OHSR Section 3.5, *General requirement* OHSR Section 3.9, *Remedy without delay* OHSR Section 4.3, *Safe machinery and equipment*
	• Logs or chunks of logs sliding or rolling off the side of a log transporter while in transit	• Ensuring at least two binders are used to secure the logs regardless of the height of the load	OHSR Section 26.68(3), *Binders*

Phase	Hazard	Control	Legislation
	• Having inadequate cab guards (aluminum structure and anchor points) • Cab guards provide the driver with life space in the event of a collision or sudden stop and are a critical safety element for log transport	• Cab guards must meet the requirements of the regulation • Frequent inspections of bull board/cab guard for damage and cracking at the anchor points • Manufacturer's instructions • P.Eng. documents to certify load rating and main supporting structure is made of steel • Certification by a P.Eng. that the main supporting structure is designed and constructed so that vibration and distortion generated by use cannot damage cab guard • If full aluminum or hybrid (blend of steel and aluminum) load rating certification for vibration and distortion required	OHSR 26.65, *Cab guard* OHS Guideline G-26.65(2)(e), *Installing the cab guard on the log transporter in a manner acceptable to WorkSafeBC*
Note: The off highway bull boards are now considered cab guards under the new regulation (definition)			

Phase	Hazard	Control	Legislation
Unloading – Self Loading Logging Truck	• Drivers working alone and they are in equipment that is unguarded • Loss of stability of the tractor	• Person check system.	OHSR Section 4.21, *Procedures for checking well-being of worker*
		• Safe work procedures (guarding)	WCA Section 21, General duties of employers OHSR Section 26.13.1(1), *Equipment operator protections*
		• Safe guards on the loading equipment (boom deflectors, etc.)	OHSR Section 26.13.1(2), *Equipment operator protections*
		• Log deck piles should be safe for the self-loading operators (i.e. should not exceed the height of the operators feet)	OHSR Section 26.2, *Planning and conducting a forestry operation*
		• Ensuring outriggers are placed as per manufacturer's instructions and that they are well maintained and safe for use	OHSR Section 4.3, Safe *machinery and equipment*

Phase	Hazard	Control	Legislation
Unloading – Conventional Logging Truck	• Removing binders prior to the unloading the load (binder removal stations)	• A load of logs must be restrained prior to the removal of binders	OHSR Section 26.69(4)(b), *Binder removal*
	• Exposure to mobile equipment	• Driver orientation to the workplace safety rules of the dry land sort or mill yard	OHSR Section 26.5, *Initial safety meeting*
		• High visibility PPE	OHSR Section 26.7, *Highly visible clothing*
	• Suspended logs being passed over the driver and/or the driver's cab	• Safe work procedures and safe landing procedures	WCA Section 21, *General duties of employers* OHSR Section 26.59, *Suspended logs*
	• Driver exposed to moving logs.	• Drivers must not be exposed to moving logs during unloading	OHSR Section 26.69(5)
	• Unsafe use of a binder removal station exposing drivers to unsecured loads	• Safe work procedures • Posted written procedures • Station must be certified by professional engineer	OHSR Section 26.69, *Binder removal*
	• Unsecured trailers	• Empty trailers must be secured when loaded onto tractors	OHSR Section 26.76, *Securing trailers*

Phase	Hazard	Control	Legislation
Reloading Trailers	• Suspended log truck trailer falling or becoming unstable	• Safe work procedures for mobile equipment	OHSR Section 26.2, *Planning and conducting a forestry operation* WCA Section 21, *General duties of employers*
		• Annual inspections for log transporter hoists	OHSR Section 14.14(g.1), *Inspection and maintenance records*
Truck Maintenance	• Equipment not locked out during maintenance activities	• Employer to ensure proper lockout procedures and safe work procedures are in place for maintenance activities on logging trucks • Supervisors and workers to ensure lock out procedures are effective and are followed at all times	WCA Section 21, *General duties of employers*
	• Tire explosions	• Following regulatory requirements for servicing multi piece rims	OHSR Section 16.47, *Training* OHSR Section 16.48, *Equipment and procedures*

Typical Orders

Order Name	OHS Regulation	Issue when…
Load Specification	OHSR Section 26.67(1.3), *Load specifications*	• Observations of logs falling off of log trucks while in transit
Load Specification (log rows)	OHSR Section 26.67(3)(b), *Load specifications*	• Observation of a compensating reach type trailer where the loaded logs are less than 12 inches from the front and rear bunks and stakes
Binders	OHSR Section 26.68, *Binders*	• Observations of a loaded logging truck where the load is above the stakes and being transported without binders — Note: Log transporters are required to use binders to secure all log loads on the highway whether above or below the stakes
Binder removal	OHSR Section 26.69(4), *Binder removal*	• Observations determine a worker or driver is removing binders without the load being restrained
Daily log	OHSR Section 26.71.2(2), *Daily log*	• Observations during an inspection of a log transport driver reveal the individual does not have a daily log or has an incomplete daily log
Warning devices	OHSR Section 26.72, *Warning devices*	• Observations in a log landing determine that a log truck was backing into a loader with no distinct audible device
Inadequate cab guard	OHSR Section 26.65, *Cab guard*	• No certification or not mounted properly
Cab guard	OHSR Section 26.65(1)(b), *Cab guard*	• Observations of a log transporter where the bull board or cab guard is not as wide as the cab

Order Name	OHS Regulation	Issue when…
Damaged cab guard	OHSR Section 26.65.6, *Cab guard*	• Observations of any damage to the cab guard such as, cracking/missing or broken pieces or corrosion
Bunks and Stakes	OHSR Section 26.66(7)(a), *Bunks and stakes*	• Observations determine that stake lines are being constructed with swaged wire rope

Officer Safety

Preparation

- Bring and wear all appropriate PPE.
  - When climbing under the truck, wear safety eyewear.
- Ensure you have your two way radio and/or satellite phone on your person at all times.

Arrival

- Check in.
- Arrange for orientation to the cut block and/or dry land sort from worker or employer rep.
- Arrange for a worker or employer rep to accompany you during the inspection.

Walk through

- Obey all signage and workplace rules.
- Never approach any logging trucks without contacting the driver.
- Prior to a detailed inspection of a logging truck, ensure all loading activities have been stopped and the truck is shut down and secured for safe inspection.
  - Truck keys are in the officer's pocket.
  - Wheel chocks are in place.
- Avoid inspecting log trucks while they are being unloaded.
- Do not conduct an inspection of a loaded logging truck unless the load is secure (i.e. binders).
- Never approach a logging truck that is being loaded on a steep slope, always keep a safe distance.
- Never try and overcome or flag down a logging truck (no jurisdiction to do so).
- Never ride in a logging truck.

Wrap up

- Communicate with employer/worker representative that the inspection is concluded and conduct exit meeting.
- Check out with employer.

Additional Resources for Employers

Name of Resource	When to direct an employer to this resource
OHS Guideline G26.2-2, *Planning log hauling operations for varying road grades*	Consultation/inspection
Ferric Study on Steep Slope Hauling	Consultation/inspection
WorkSafeBC Video on Log Landing Safety for Truck Drivers	Consultation/safety meeting
Truck Safe – BC Forest Safety Council	Consultation/inspection
Log Transporter ID Chart – Terry	Consultation/inspection
Operator's Manual and Manufacturer's Instructions	Consultation/inspection
U bolt video	Consultation/inspection
FPInnovations over exertion injuries reduction	Consultation/injury reduction http://teamsites/sites/ts_industries/forestry/Activities/FP%20Innovations_LoadWrapper.pdf https://www.youtube.com/watch?v=Jnbh–ZVNC5U
Truck Steep Slope Calculator – FPInnovations	Consultation/inspection
Professional Log Truck Driver Competency Program – BC Forest Safety Council	Consultation/inspection